

Speculative Decoding for Hybrid SSM–Attention Models: A Case Study with PLaMo 2 on llama.cpp

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Abstract

Speculative decoding is a standard technique for accelerating autoregressive LLM inference, but it is poorly supported for hybrid state-space-model (SSM) architectures: the fixed-size recurrent state of Mamba-style layers cannot be truncated on token rejection the way a KV cache can, and vLLM currently rejects all speculative-decoding methods on Mamba-hybrid models outright. llama.cpp recently added a generic SSM-state checkpoint/rollback mechanism, making it—to our knowledge—the first practical platform to test speculative decoding on such models. We benchmark **pfnet/plamo-2-8b** (target) with **pfnet/plamo-2-1b** (draft) on an Apple M4 Pro (Metal backend, llama.cpp build b9596) across three task families with greedy sampling, verifying byte-identical outputs versus the non-speculative baseline. The result is a carefully measured *negative* one: despite token acceptance rates up to 90% and a draft model that is $6.3\times$ faster standalone, draft-model speculation is never profitable on this hardware (best case $0.99\times$; worst $0.29\times$). Only model-free n-gram (prompt-lookup) drafting yields a speedup, and only on translation ($1.21\times$). We attribute the slowdown to per-round costs—serial draft decoding, batched verification, and ~ 33.5 MiB SSM-state checkpoints—that exceed the savings on bandwidth-bound consumer hardware. Along the way we discovered a novel quantization bug: quantizing PLaMo 2’s Mamba output projection (**ssm_out**) to Q8_0 causes degenerate generation on newline-containing prompts in both the 1B and 8B models; we isolate the faulty tensor experimentally and provide a one-flag mitigation that likely applies to all community PLaMo 2 GGUFs.

1 Introduction & Motivation

PLaMo 2 is Preferred Networks’ (PFN) family of Japanese/English foundation models built on a Samba-style hybrid architecture that interleaves Mamba2 state-space (SSM) layers with sliding-window attention [1]. Hybrid SSM–attention models are attractive for inference because the SSM layers replace an ever-growing KV cache with a fixed-size recurrent state, but that same property breaks one of the most widely used inference accelerations: speculative decoding.

Speculative decoding drafts k candidate tokens cheaply (with a small draft model, or model-free heuristics such as prompt lookup), verifies them in a single batched forward pass of the target model, and—under greedy sampling—is *lossless*: output is provably identical to plain decoding. The catch for hybrid models is rejection. With a pure transformer, rejecting speculated tokens means truncating the KV cache, an $O(1)$ bookkeeping operation. With an SSM layer, the recurrent state after processing k tokens has irreversibly mixed those tokens in; there is no truncation operator. As of this writing, vLLM raises a `NotImplementedError` (“Mamba with speculative decoding is not supported yet”) for *all* speculative methods on Mamba-hybrid models [2].

llama.cpp recently solved rollback generically via per-sequence rolling checkpoints of SSM state [3], opening a window to ask an empirical question that, to our knowledge, has not been answered for PLaMo 2: *does speculative decoding actually pay off on a hybrid SSM–attention model, once correctness is solved?* We answer it for the PLaMo 2 8B/1B pair on Apple Silicon, with a rigorous losslessness check, and report an honest negative result with a mechanistic explanation. As a secondary, first-class contribution, we document and root-cause a PLaMo 2 quantization bug that silently corrupts generation in standard Q8_0 GGUFs.

2 Background

2.1 Samba-style hybrid architecture

PLaMo 2 follows the Samba recipe: blocks of Mamba2 SSM layers interleaved with sliding-window attention layers [1]. The SSM layers carry a fixed-size recurrent state (convolution state plus SSM state) per sequence; the attention layers keep a bounded sliding-window KV cache. This gives near-constant memory during decode, at the cost of state that evolves destructively token by token.

2.2 KV truncation vs. SSM state rollback

In a transformer, speculative verification that accepts $a < k$ drafted tokens simply discards KV entries for positions beyond a . An SSM has no analogous operation: its state is a lossy summary of the whole prefix, so “forgetting” the last $k - a$ tokens requires having saved an earlier copy of the state. This is exactly why vLLM’s speculative-decoding path refuses Mamba-hybrid models for every proposer type (n-gram, draft model, EAGLE, Medusa, MTP) [2].

2.3 llama.cpp’s recurrent checkpoint mechanism

llama.cpp addressed this generically in its recurrent memory module: `llama_memory_recurrent` maintains a rolling buffer of checkpoints (depth 8 per sequence) of the SSM state tensors, saved at batch boundaries and restored when speculation is rejected [3]. Correctness becomes architecture-agnostic, but the checkpoints are not free: for the PLaMo 2 8B, server logs in our runs reported approximately 33.5 MiB per context checkpoint. Every speculation round therefore adds state-copy traffic on top of draft decoding and batched verification—a cost model that turns out to matter a great deal in Section 6.

3 A Quantization Bug Found Along the Way

Before any speculation numbers could be trusted, we hit a correctness bug that we believe affects all community PLaMo 2 GGUFs and present as a first-class contribution.

Symptom. In the first benchmark run, every prompt containing a bare newline (token 10, the reserved byte token `0x0A`)—i.e. all code and translation prompts—returned *empty* content while “generating” 256 tokens at full speed. This occurred with both the 1B and the 8B Q8_0 models.

Root-cause chain. Each step below is one controlled experiment:

1. Requesting raw token IDs (`return_tokens: true`) revealed that the model emits token 31, i.e. `<|plamo:reserved:0x1F|>`, indefinitely; reserved tokens render as empty text and are not end-of-generation, so decoding runs to the limit.
2. Tokenizer round-trip via `/tokenize + /detokenize` is perfect, including indented code. Not a tokenizer bug.
3. Full prompt token-ID comparison, HF transformers vs. llama.cpp: identical (modulo BOS; manually prepending BOS changed nothing).
4. Ground truth via HF transformers on CPU in fp32 (this required pinning `transformers<4.58` and writing pure-PyTorch shims for the `causal_conv1d` and `mamba_ssm` reference functions, since pfnut’s modeling code’s CPU fallback still calls those packages): the reference implementation generates correct code for the failing prompt. Hence a llama.cpp-side problem.
5. Unquantized bf16 GGUF in llama.cpp: correct output. Hence a quantization problem, not an inference-kernel problem.
6. Re-quantizing to Q8_0 from bf16 with `llama-quantize`: still broken. Hence not a converter bug.

7. Per-tensor isolation with `--tensor-type X=bf16` (Table 1): keeping `ssm_out` alone in bf16 fixes generation; exempting any other tensor does not.

Table 1: Per-tensor isolation: which single tensor kept in bf16 (all else Q8_0) repairs generation on newline-containing prompts.

Tensor(s) kept in bf16	Role	Output
token embeddings	embedding	broken
output	LM head	broken
all <code>ssm_*</code> matrices	all Mamba projections	fixed
<code>ssm_in</code>	Mamba input projection	broken
<code>ssm_x</code>	Mamba x projection	broken
<code>ssm_dt</code>	Mamba Δt projection	broken
<code>ssm_out</code>	Mamba output projection	fixed

Conclusion and mitigation. Quantizing PLaMo 2’s Mamba output projection (`ssm_out`) to Q8_0 causes degenerate generation on newline-containing prompts. The mitigation is one flag:

```
llama-quantize --tensor-type ssm_out=bf16 model-BF16.gguf model-Q8fixed.gguf Q8_0
```

Impact. The bug reproduces on both the 1B and 8B models and is a property of the standard Q8_0 recipe, so it likely affects all community PLaMo 2 GGUFs published to date. A natural upstream fix is to protect `ssm_out` in llama.cpp’s plamo2 quantization recipe. All benchmarks in this paper use the repaired quantization (“Q8fixed”: Q8_0 with `ssm_out` in bf16) for *both* models.

4 Methodology

Hardware and build. MacBook Pro with Apple M4 Pro, 48 GB unified memory, Metal backend. llama.cpp build b9596 (official release binaries), served via `llama-server`.

Models and quantization. Target: `pfnet/plamo-2-8b`. Draft: `pfnet/plamo-2-1b`. Both converted to GGUF and quantized to Q8_0 with `ssm_out` kept in bf16 (Section 3). Standalone, the draft decodes at 134 tok/s vs. 21.4 tok/s for the target—a 6.3 $\times$ ratio that would normally make draft-model speculation attractive.

Configurations.

- `fx_baseline`: plain decoding, no speculation.
- `fx_spec_d{2,4,8,16}`: draft-model speculation (`--spec-type draft-simple`) with `draft-n-max = 2,4,8,16` and `draft-n-min = 1`.
- `fx_ngram`: model-free n-gram speculation (`--spec-type ngram-simple`, prompt lookup; no draft model).

Tasks and protocol. Twelve prompts spanning three task families —`ja_chat`, `translation_en_ja`, and `code`—each run 3 times per configuration (`bench/prompts.json`). Greedy sampling (temperature 0, top- k 1), `n_predict` 256. We record decode throughput, drafted/accepted token counts, and a SHA-256 hash of the full generated text per prompt.

Losslessness verification. For every prompt, the SHA-256 of the speculative output is compared against the baseline output. All speculative configurations produced byte-identical greedy output to the baseline, confirming that llama.cpp’s SSM checkpoint/rollback is exact.

5 Results

Table 2 reports median decode throughput per task, speedup versus baseline, and token acceptance rate. The headline: with a real draft model, speculation never wins on this hardware. The best draft-model cell is `fx_spec_d2` on translation at $0.99\times$ (break-even); `ja_chat` degrades to $0.29\times$ at depth 16. The only profitable configuration anywhere is n-gram drafting on translation ($1.21\times$), with `fx_spec_d4` on translation near-breakeven-positive at $1.18\times$.

Table 2: Median decode throughput in tok/s (speedup vs. baseline) [token acceptance rate]. Greedy sampling; all speculative outputs byte-identical to baseline.

Config	code	ja_chat	translation_en_ja
<code>fx_baseline</code>	19.7 (1.00 $\times$)	21.6 (1.00 $\times$)	15.7 (1.00 $\times$)
<code>fx_ngram</code>	18.1 (0.92 $\times$) [19%]	17.6 (0.81 $\times$) [63%]	19.0 (1.21$\times$) [23%]
<code>fx_spec_d2</code>	16.5 (0.84 $\times$) [90%]	13.8 (0.64 $\times$) [80%]	15.5 (0.99 $\times$) [87%]
<code>fx_spec_d4</code>	12.0 (0.61 $\times$) [77%]	10.2 (0.47 $\times$) [55%]	18.5 (1.18 $\times$) [70%]
<code>fx_spec_d8</code>	11.6 (0.59 $\times$) [55%]	7.9 (0.36 $\times$) [32%]	11.1 (0.71 $\times$) [48%]
<code>fx_spec_d16</code>	11.7 (0.59 $\times$) [33%]	6.2 (0.29 $\times$) [17%]	9.3 (0.59 $\times$) [29%]

Figure 1 summarizes speedups per task and configuration; Figure 2 shows acceptance rates, which decay sharply with draft depth; Figure 3 shows full throughput distributions across prompts and repeats; Figure 4 relates acceptance to realized speed, making the central paradox visible: 90% acceptance still loses.

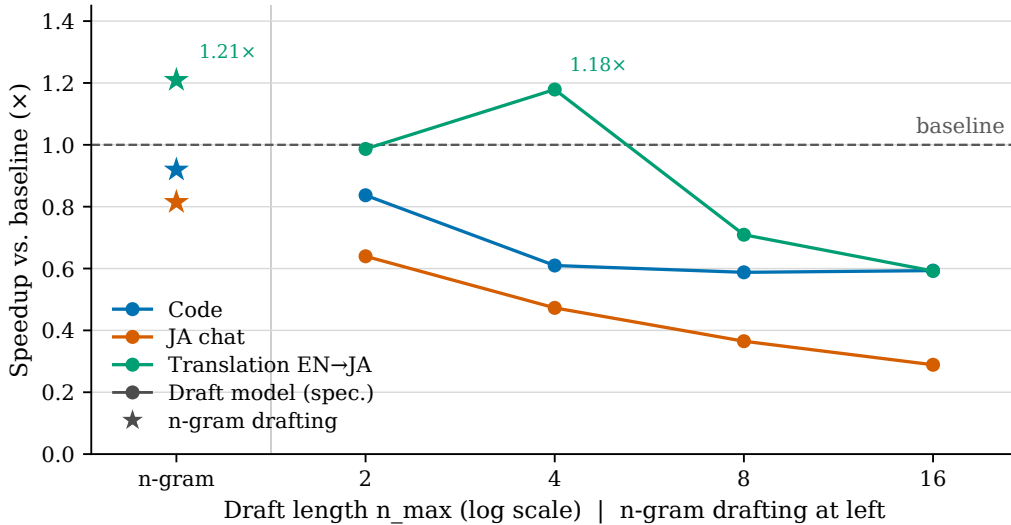


Figure 1: Decode speedup vs. baseline by task and configuration. Only n-gram drafting on translation exceeds $1.0\times$; draft-model speculation is unprofitable everywhere.

Losslessness. All speculative configurations produced byte-identical greedy output to baseline, verified by SHA-256 per prompt. The negative result is therefore purely about throughput, not correctness.

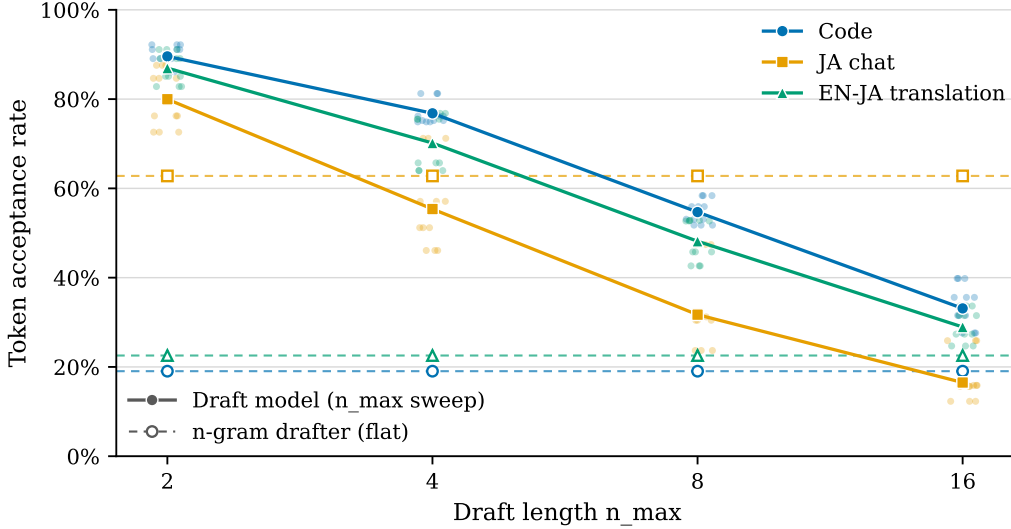


Figure 2: Token acceptance rate by task and draft depth. Acceptance is excellent at depth 2 (up to 90%) but per-token acceptance decays with depth, so deeper drafts amortize worse.

6 Discussion

Why 90% acceptance still loses. Each speculation round pays three costs that plain decoding does not: (i) *serial draft-model decode*—the 1B draft, though $6.3\times$ faster standalone, still runs serially before each verification and competes for the same unified-memory bandwidth; (ii) *batched verification* in the target, which on bandwidth-bound Metal hardware is far from free relative to single-token decode (single-token decode already saturates memory bandwidth, so a k -token verify costs nearly as much per token as plain decode); and (iii) *SSM-state checkpointing*—roughly 33.5 MiB saved per context checkpoint for the 8B, plus restores on rejection. At depth 2 with 90% acceptance (code), these round costs already eat the entire benefit ($0.84\times$). Deeper drafts amortize per-round costs over more tokens in principle, but per-token acceptance decays with depth (Figure 2), so realized acceptance lengths grow slowly while rejected work and rollbacks grow quickly: depth 16 lands at $0.29\text{--}0.59\times$.

Why n-gram wins on translation. Prompt-lookup drafting has essentially zero proposal cost—no draft forward pass, no second model resident in memory. Translation outputs copy long spans from the source prompt verbatim (names, numbers, code-switched terms), so even a modest 23% acceptance converts into free accepted tokens, yielding the study’s only speedup ($1.21\times$). On code, ja_chat, and any task where output text is not substring-recoverable from the prompt, lookup proposals miss and the verification overhead makes it a net loss.

Implications. For vLLM, these numbers temper the urgency of lifting the Mamba-hybrid speculative-decoding restriction [2]: correctness machinery (as llama.cpp shows) is solvable, but a separate draft model may simply not pay on bandwidth-bound hardware, and the checkpoint traffic is an architecture-intrinsic tax. Cheap proposers (n-gram, and especially self-drafting methods such as MTP or EAGLE-style heads that reuse the target’s hidden states and add no second model) are the more promising direction for hybrids, since they attack the dominant cost—the serial draft—directly. Notably, PLaMo 3 NICT models move to an attention-only architecture, which sidesteps SSM rollback entirely and restores cheap KV truncation; speculative decoding economics there should look conventional again.

Framing. We deliberately report this as a measured negative result with a mechanism. “Speculation is lossless and works, but is unprofitable for PLaMo 2 8B/1B on M4 Pro Metal” is directly actionable for

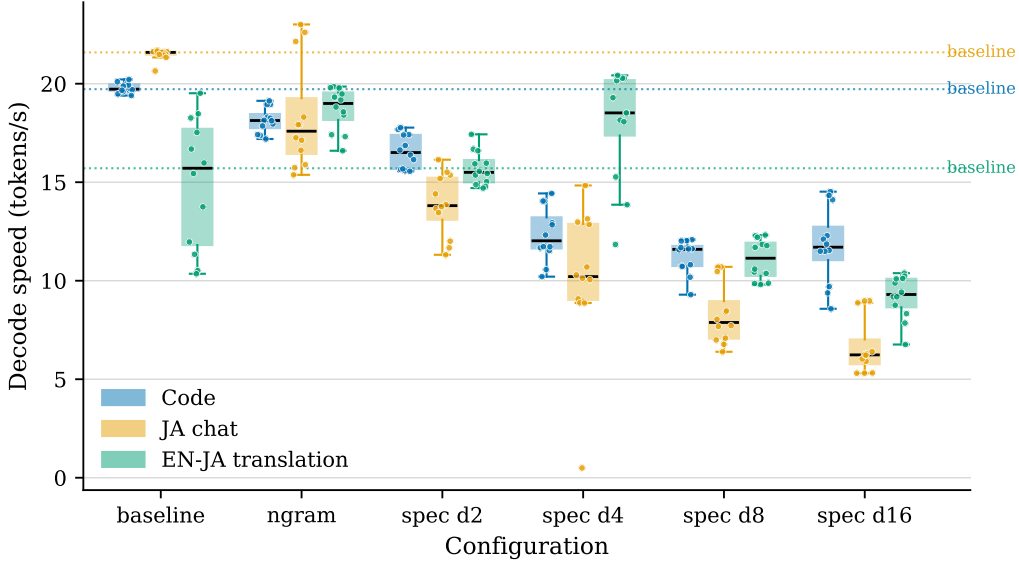


Figure 3: Decode throughput distributions across prompts and repeats for each configuration and task.

anyone deploying these models locally: run the baseline, or n-gram for translation-like workloads.

7 Limitations

- **Single hardware point.** All measurements are on one M4 Pro with Metal. CUDA GPUs with higher compute-to-bandwidth ratios and cheaper batched verification could flip the economics; we have no CUDA leg yet.
- **Small prompt suite.** 12 prompts $\times$ 3 repeats across three tasks captures task-level trends but not workload diversity (long-context, RAG, agentic loops with heavy prompt reuse—likely favorable to n-gram drafting).
- **One model pair and one quantization.** Results are for PLaMo 2 8B/1B at Q8fixed; other draft sizes, quantizations, or speculation parameters (e.g. adaptive depth) were not swept.

8 Reproducibility

All artifacts live under `/Users/gurunathlunkupalivenugopal/ionet/pfn-try`:

- `bench/prompts.json` — 12 prompts across the three tasks.
- `results/fx_baseline.jsonl`, `results/fx_spec_d{2,4,8,16}.jsonl`, and `results/fx_ngram.jsonl` — one JSON object per line with fields `label`, `task`, `prompt_idx`, `rep`, `tok_per_sec`, `n_predicted`, `draft_n`, `draft_n_accepted`, `prompt_per_sec`, `content_sha`, `content`.
- `report/figures/` — figure PDFs plus the Python scripts that generate them.
- `commands-log.md` — the full command log, including the `ssm_out` bug-hunt chain (section 3) and harness fixes (section 4).

Key commands:

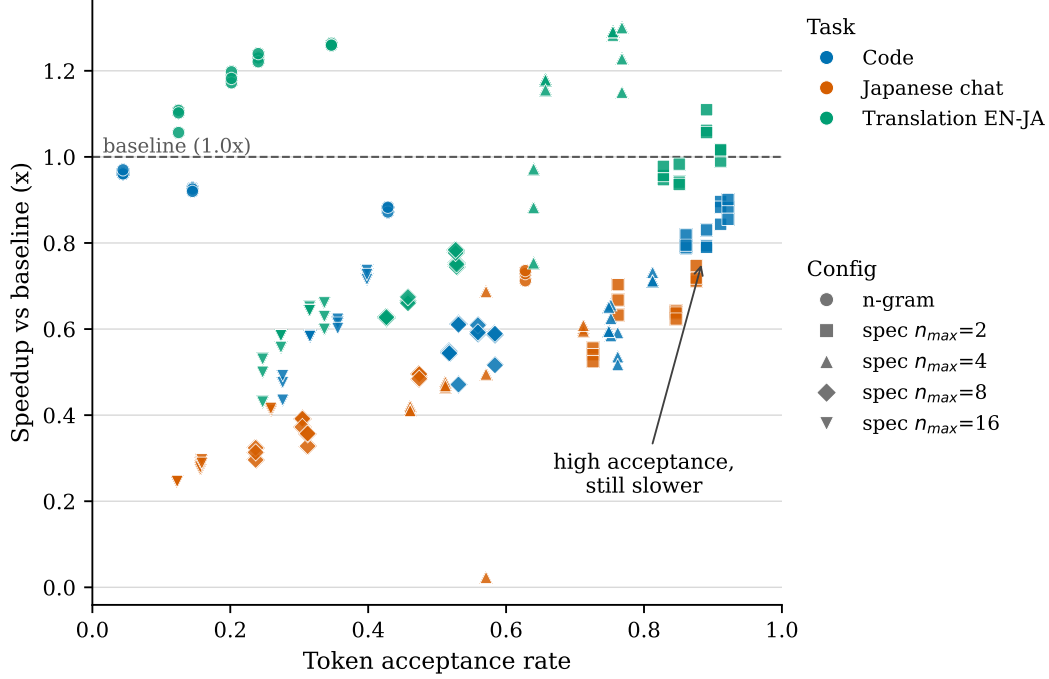


Figure 4: Acceptance rate vs. realized decode speed. High acceptance does not translate into speedup: per-round overheads dominate.

```
llama-quantize --tensor-type ssm_out=bf16 plamo-2-8b-BF16.gguf plamo-2-8b-Q8fixed.gguf Q8_0
llama-server -m plamo-2-8b-Q8fixed.gguf -md plamo-2-1b-Q8fixed.gguf --spec-type draft-simple
--draft-max {2|4|8|16} --draft-min 1
llama-server -m plamo-2-8b-Q8fixed.gguf --spec-type ngram-simple
```

Two harness pitfalls worth repeating: on build b9596, `-md` alone loads the draft model but leaves speculation *off* (the log prints “no implementations specified for speculative decoding”); `--spec-type draft-simple` is required, and engagement should be confirmed via the `timings.draft_n` / `draft_n_accepted` fields in responses. Greedy losslessness should be checked with a real content hash (SHA-256), not Python’s per-process-salted `hash()`.

References

- [1] Preferred Networks. *PLaMo 2 Technical Report*. arXiv:2509.04897, 2025. <https://arxiv.org/abs/2509.04897>
- [2] vLLM project. “Mamba with speculative decoding is not supported yet.” GitHub issue #30114, vllm-project/vllm. <https://github.com/vllm-project/vllm/issues/30114>
- [3] ggml-org/llama.cpp. Recurrent state checkpointing for speculative decoding (`llama_memory_recurrent` rolling checkpoint buffer, depth 8 per sequence). Pull request #20075. <https://github.com/ggml-org/llama.cpp/pull/20075>